```
FUNCTION displayMenu()
    PRINT "=== University Library Management System ==="
    PRINT "1. Member Registration"
    PRINT "2. Display Available Books"
    PRINT "3. Borrow a Book"
    PRINT "4. Return a Book"
    PRINT "5. Display Member Record"
    PRINT "6. Exit"
    PRINT "Enter your choice:"
ENDFUNCTION.

FUNCTION registerMember(name, studentID*, memberType*)
    INPUT name, studentID
    INPUT
    PRINT "Enter member name:"
    INPUT name
    PRINT "Enter member ID:"
    INPUT studentID.

    DO
      PRINT "Enter member type (1:Student, 2: Faculty):"
      INPUT value
      *memberType = value
```

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```
IF *memberType != 1 AND
   *memberType != 2 THEN
   PRINT "Invalid type. Please enter 1 or 2."
ENDIF
WHILE *memberType != 1 AND *memberType != 2

PRINT "Registration successful."
ENDFUNCTION.

FUNCTION displayBooks(bookIDs[], bookTitles[],
                      availableCopies[], totalBooks).
    PRINT "ID | Title | Copies"
    FOR i = 0 TO totalBooks - 1
        PRINT bookIDs[i] + " | " + bookTitles[i]
              + " | " + availableCopies[i].
    NEXT i
ENDFUNCTION.

FUNCTION borrowBook(bookIDs[], bookTitles[],
    availableCopies[], totalBooks, borrowedBookIDs[],
    borrowedCount*, memberType).
  IF *borrowedCount >= 3 THEN
    PRINT "Error: Maximum borrowing limit
           (3 books) reached."
    RETURN
  ENDIF
```

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```
PRINT "Enter book ID to borrow: "
INPUT reqID

DO
    PRINT "Enter intended number of days to
           keep: "
    INPUT reqDays
    IF reqDays <=0 THEN
        PRINT "Days must be positive."
    ENDIF
WHILE reqDays <= 0

index = -1
FOR i = 0 TO totalBooks - 1
    IF bookIDs[i] == reqID THEN
        index = i
        BREAK
    ENDIF
NEXT i

IF index == -1 THEN
    PRINT "Error: Invalid book ID."
ELSE IF availableCopies[index] == 0 THEN
    PRINT "Error: No copies of this book are
           currently available."
ELSE
    availableCopies[index] = availableCopies[index] - 1
```

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```
        borrowedBookIDs[*borrowedCount] = reqID
        *borrowedCount = *borrowedCount + 1
        PRINT "Successfully borrowed book."
    ENDIF
ENDFUNCTION.

FUNCTION returnBook(bookIDs[], availableCopies[],
                    totalBooks, borrowedBookIDs[],
                    borrowedCount*, memberType, totalFine*)
    PRINT "Enter book ID to return:"
    INPUT retID

    borrowedIndex = -1
    FOR i = 0 TO *borrowedCount - 1
        IF borrowedBookIDs[i] == retID THEN
            borrowedIndex = i
            BREAK
        ENDIF
    NEXT i

    IF borrowedIndex == -1 THEN
        PRINT "Error: You have not borrowed
               this book."
        RETURN 0.0
    ENDIF
```

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```
DO
    PRINT "Enter actual number of days kept:"
    INPUT actualDays
    IF actualDays <> 0 THEN
        PRINT "Days must be positive."
    ENDIF
WHILE actualDays <= 0

IF memberType == 1 THEN
    allowedDays = 14
    finalRate = 20
ELSE
    allowedDays = 30
    finalRate = 10
ENDIF

lateDays = actualDays - allowedDays
currentFine = 0.0

IF lateDays > 0 THEN
    currentFine = lateDays * fineRate
ENDIF

*totalFine = *totalFine + currentFine
payableFine = currentFine.
```

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```
IF currentFine > 0 THEN
    PRINT "Calculated Fine: Rs." + currentFine
          + ". Enter membership code for
          waiver (or 'none'): "
    INPUT code
    IF code == "LIB2026" THEN
        payableFine = currentFine * 0.50
        PRINT "Code accepted! 50% waiver
                applied."
    ELSE
        PRINT "Invalid code. No waiver applied."
    ENDIF
ENDIF

FOR i = 0 TO totalBooks - 1
    IF bookIDs[i] == retID THEN
        availableCopies[i] = availableCopies[i]
                             + 1
        BREAK
    ENDIF
NEXT i

FOR i = borrowedIndex TO *borrowedCount - 2
    borrowedBookIDs[i] = borrowedBookIDs[i+1]
NEXT i

*borrowedCount = *borrowedCount - 1
PRINT "Successfully returned book. Fine payable:
```


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```
        Rs." + payableFine
        RETURN payableFine
ENDFUNCTION.

FUNCTION displayMemberRecord(name, studentID,
                memberType, borrowedBookIDs[],
                borrowedCount, totalFine,
    PRINT "=== Member Record ==="
    PRINT "Name: " + name
    PRINT "ID: " + studentID
    IF memberType == 1 THEN
        PRINT "Type: Student"
    ELSE
        PRINT "Type: Faculty"
    ENDIF

    PRINT "Currently Borrowed Book IDs:"
    IF borrowedCount == 0 THEN
        PRINT "None"
    ELSE
        FOR i = 0 TO borrowedCount - 1
            PRINT borrowedBookIDs[i] + " "
        NEXT i
    ENDIF
    PRINT "\n"

    PRINT "Total fine charged (before waiver): Rs."
        + totalFine.
```

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PRINT "Final fine payable (after waiver): Rs. "
+ fineAfterWaiver.
END FUNCTION.

BEGIN main
bookIDs = [101, 102, 103, 104]
bookTitles = ["Programming in C", "Data Structures",
"Database Systems", "Computer Networks"].
availableCopies = [5, 3, 0, 4]
totalBooks = 4

name = "Not Registered"
studentID = 0
memberType = 0

DECLARE borrowedBookIDs[3]
borrowedCount = 0

totalFine = 0.0
fineAfterWaiver = 0.0

DO
CALL displayMenu()
INPUT choice

IF choice >= 3 AND choice <= 5 AND
memberType == 0 THEN.

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```
PRINT "Please register a member first (option 1)!"
ELSE IF choice == 1 THEN
    CALL registerMember(name, ~~studentID~~,
        *studentID, *memberType).
ELSE IF
    CALL displayBooks(bookIDs, bookTitles,
        availableCopies, totalBooks).
ELSE IF choice == 3 THEN
    CALL borrowBook(bookIDs, bookTitles,
        availableCopies, totalBooks,
        borrowedBookIDs, ~~bor~~ * BorrowedCount,
        memberType).
ELSE IF choice == 4 THEN
    payable = CALL returnBook(bookIDs,
        availableCopies, totalBooks,
        borrowedBookIDs, *borrowed
        Count, memberType, * totalFine)
    ~~Final~~fineAfterWaiver = fineAfterWaiver + payable
ELSE IF choice == 5 THEN
    CALL displayMemberRecord(name, studentID,
        memberType, borrowedBookIDs,
        borrowedCount, totalFine,
        fineAfterWaiver).
ELSE IF choice == 6 THEN
    PRINT "Exiting Library System. Goodbye!
ELSE
    PRINT "Invalid option. Please try again."
ENDIF
```

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WHILE choice != 6

END.

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